Mathematics 8 — Solved Exercises

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Chapters covered in this notes PDF: Domain 1 — Numbers and Operations, Sub-Domains (i) Real Numbers, (ii) Estimation and Approximation, (iii) Square Roots and Cube Roots: Exercises 1.1 to 1.11 and Review Exercise 1 (a) (book pp. 1-33). PARTIAL — Sub-Domains (iv) Sets onwards are not yet included; they will be added in a later pass.

Domain 1 — Numbers and Operations: Real Numbers

Exercise 1.1

Q1. Identify which of the following fractions are terminating and which are recurring decimal fractions?

Rule: First write each fraction in its lowest terms. If the denominator has only 2s and 5s as factors, the decimal terminates.

If the denominator has any other prime factor, the decimal recurs (repeats).

Terminating: (iv) 2/5 = 0.4; (viii) 6/10 = 3/5 = 0.6; (ix) 16/20 = 4/5 = 0.8; (x) 9/10 = 0.9;

(xi) 1/8 = 0.125; (xii) 14/20 = 7/10 = 0.7; (xiv) 4/5 = 0.8; (xv) 1/4 = 0.25; (xvii) 8/10 = 4/5 = 0.8; (xviii) 3/5 = 0.6.

Recurring: (i) 4/6 = 2/3 = 0.333...; (ii) 11/12, 12 = 2 × 2 × 3, so 0.9166...;

(iii) 1/3 = 0.333...; (v) 8/11 = 0.7272...; (vi) 5/12 = 0.4166...; (vii) 5/7 = 0.714285...;

(xiii) 2/3 = 0.666...; (xvi) 7/9 = 0.777....

Q2. Separate rational and irrational numbers from the following:

Rational numbers can be written as p/q, or as terminating or recurring decimals. Irrational numbers cannot.

Rational: (ii) 1/7; (iii) 2.152 (terminating); (iv) 0.818281828182... (the block 82 repeats, so recurring);

(v) 2.751 (terminating); (vi) 22/7; (viii) √25 = 5.

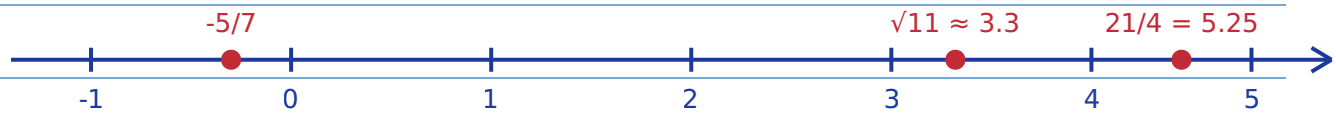
Irrational: (i) √5, because 5 is not a perfect square; (vii) √14, because 14 is not a perfect square.

Q3. Represent the following numbers on the number line: (i) 21/4, (ii) -53/3, (iii) √11, (iv) 25/2, (v) -5/7, (vi) √31.

Step 1: Change each number to a decimal to find its place: 21/4 = 5.25; -53/3 = -17.67 (approx.);

√11 = 3.32 (approx.); 25/2 = 12.5; -5/7 = -0.71 (approx.); √31 = 5.57 (approx.).

Step 2: Draw a number line with equal gaps and mark 0, then count to each position, as in the diagram below.



Step 3: Mark -5/7 between -1 and 0, close to -1; mark √11 between 3 and 4, close to 3; mark 21/4 between 5 and 6;

mark √31 between 5 and 6, a little past the middle; mark 25/2 at 12.5, midway between 12 and 13;

and mark -53/3 between -18 and -17, close to -18. Each point sits exactly at its decimal value.

Q4. Find the absolute value of each of the following: (i) 15, (ii) -21, (iii) -36, (iv) -9, (v) -100, (vi) 88.

Rule: The absolute value |x| is the distance of x from 0, so it is never negative: |x| = x if x is positive, and -x if x is negative.

(i) |15| = 15. (ii) |-21| = 21. (iii) |-36| = 36.

(iv) |-9| = 9. (v) |-100| = 100. (vi) |88| = 88.

Q5. Find the absolute difference between each of the following: (i) -7 and -2, (ii) 8 and -3, (iii) -5 and 1, (iv) 2 and 3.

Rule: The absolute difference of a and b is |a - b|, the distance between them on the number line.

(i) |-7 - (-2)| = |-7 + 2| = |-5| = 5.

(ii) |8 - (-3)| = |8 + 3| = |11| = 11.

(iii) |-5 - 1| = |-6| = 6.

(iv) |2 - 3| = |-1| = 1.

Q6. 50 3/4 kilograms of rice is to be packed equally in 5 1/2 kilogram packets. Find: (a) how many packets of rice will be packed? (b) how much rice will be there in 8 packets of mass 5 1/2 kilograms?

Step 1: Total rice = 50 3/4 = 203/4 kg. One packet = 5 1/2 = 11/2 kg.

Step 2: (a) Number of packets = 203/4 ÷ 11/2 = 203/4 × 2/11 = 406/44 = 203/22 = 9.23 (approx.).

Step 3: So 9 full packets can be packed, using 9 × 11/2 = 99/2 = 49.5 kg of rice.

Step 4: (b) Rice in 8 packets = 8 × 11/2 = 88/2 = 44 kg.

Q7. To decorate a birthday party, Zara used ribbons of two colours. The length of blue ribbon is 15 2/3 metres. The length of green ribbon is 4/5 times the length of blue ribbon. (a) What is the length of green ribbon? (b) Find the total length of both the ribbons.

Step 1: Blue ribbon = 15 2/3 = 47/3 metres.

Step 2: (a) Green ribbon = 4/5 × 47/3 = 188/15 = 12.53 metres (approx.).

Step 3: (b) Total length = 47/3 + 188/15 = 235/15 + 188/15 = 423/15 = 141/5 = 28.2 metres.

Q8. If 230.75 kilograms of jaggery powder is to be packed equally in packets, each packet contains 2.5 kg of jaggery powder. Find the number of required packets. Also find how much jaggery powder will be there in 15 packets of mass 2.5 kg.

Step 1: Number of packets = total mass ÷ mass of one packet = 230.75 ÷ 2.5.

Step 2: 230.75 ÷ 2.5 = 2307.5 ÷ 25 = 92.3. So 92 full packets can be packed.

Step 3: Mass in 15 packets = 15 × 2.5 = 37.5 kg.

Q9. Zeeshan wants to fill 137.5 litres of petrol in bottles. Find the number of bottles if: (a) the capacity of each bottle is 5.5 litres, (b) the capacity of each bottle is 6.25 litres, (c) also tell how much petrol will be there in 18 bottles of capacity 6.25 litres each.

Step 1: (a) Number of bottles = 137.5 ÷ 5.5 = 1375 ÷ 55 = 25 bottles.

Step 2: (b) Number of bottles = 137.5 ÷ 6.25 = 13750 ÷ 625 = 22 bottles.

Step 3: (c) Petrol in 18 bottles = 18 × 6.25 = 112.5 litres.

Exercise 1.2

Q1. Identify which property of real numbers is used in the following:

(i) 11 + 0 = 11: Additive identity, because adding 0 leaves a number unchanged.

(ii) x × (y × z) = (x × y) × z: Associative property of multiplication, because the grouping changes but the product does not.

(iii) 12 + (-12) = 0: Additive inverse, because a number plus its negative gives 0.

(iv) 14 + 20 = 20 + 14: Commutative property of addition, because the order changes but the sum does not.

(v) $-3 \times \sqrt{5} \in R$: Closure property of multiplication, because the product of two real numbers is a real number.

(vi) $2 \times 1/2 = 1$: Multiplicative inverse, because a number times its reciprocal gives 1.

(vii) $22 \times 1 = 22$: Multiplicative identity, because multiplying by 1 leaves a number unchanged.

(viii) $(-6/10) \times (-10/6) = 1$: Multiplicative inverse, because the two numbers are reciprocals of each other.

(ix) $(19 + 12) + 8 = 19 + (12 + 8)$: Associative property of addition.

(x) $-3 + (23 + 48) = (-3 + 23) + 48$: Associative property of addition.

(xi) $89 \times 1 = 89$: Multiplicative identity.

(xii) $12 \times 46 = 46 \times 12$: Commutative property of multiplication.

(xiii) $\sqrt{27} + 0 = \sqrt{27}$: Additive identity.

(xiv) $x \times y = y \times x$: Commutative property of multiplication.

(xv) $\pi + (-\pi) = 0$: Additive inverse.

Q2. Identify which order property is used in the following: (i) 15, (ii) -27, (iii) 7/9, (iv) -9/16, (v) -20/23, (vi) 1, (vii) -2/3, (viii) 7/2, (ix) -7, (x) 12.

Note: The book lists single numbers here, so each number is compared with 0 by the trichotomy property.

Trichotomy property: For every real number a, exactly one of $a > 0$, $a < 0$ or $a = 0$ is true.

Greater than 0: (i) $15 > 0$; (iii) $7/9 > 0$; (vi) $1 > 0$; (viii) $7/2 > 0$; (x) $12 > 0$.

Less than 0: (ii) $-27 < 0$; (iv) $-9/16 < 0$; (v) $-20/23 < 0$; (vii) $-2/3 < 0$; (ix) $-7 < 0$.

Q3. Identify which order property is used in the following:

(i) $x < y$ or $x > y$ or $x = y$: Trichotomy property, because exactly one of the three holds for any two real numbers.

(ii) $a > b \Rightarrow a/c < b/c$, where $c < 0$: Division property, because dividing by a negative number reverses the inequality.

(iii) $a > b \Rightarrow a - c > b - c$: Subtraction property, because subtracting the same number from both sides keeps the inequality.

(iv) $5 > 2 \Rightarrow 5 \cdot c < 2 \cdot c$, where $c < 0$: Multiplication property, because multiplying by a negative number reverses the inequality.

(v) $7 > z$ and $z > 5 \Rightarrow 7 > 5$: Transitive property, because order passes through the middle number.

(vi) $10 < 15 \Rightarrow 10 \cdot c > 15 \cdot c$, where $c < 0$: Multiplication property, because multiplying by a negative number reverses the inequality.

(vii) $x < y \Rightarrow x - w < y - w$: Subtraction property.

(viii) $x < z \Rightarrow x/\ell > z/\ell$, where $\ell < 0$: Division property, because dividing by a negative number reverses the inequality.

Domain 1 — Numbers and Operations:

Estimation and Approximation

Exercise 1.3

Q1. Round the following numbers to the given degree of accuracy: (i) 4763 (2 s.f.), (ii) 785029 (3 s.f.), (iii) 8095321 (4 s.f.), (iv) 0.0289357 (5 s.f.), (v) 2345987 (5 s.f.), (vi) 0.780247 (4 s.f.).

Rule: Look at the digit just after the required place. If it is 5 or more, add 1 to the last kept digit; if it is less than 5, leave it unchanged.

(i) 4763: the first two significant digits are 47, the next digit is 6 (5 or more), so 47 rounds up to 48, giving 4800 (2 s.f.).

(ii) 785029: the first three significant digits are 785, the next digit is 0, so it stays 785, giving 785000 (3 s.f.).

(iii) 8095321: the first four significant digits are 8095, the next digit is 3, so it stays 8095, giving 8095000 (4 s.f.).

(iv) 0.0289357: the significant digits are 289357; keeping five gives 28936 (next digit 7 rounds the 5 up to 6), so 0.028936 (5 s.f.).

(v) 2345987: the first five significant digits are 23459, the next digit is 8, so it rounds up to 23460, giving 2346000 (5 s.f.).

(vi) 0.780247: the significant digits are 780247; keeping four gives 7802 (next digit 4 leaves it unchanged), so 0.7802 (4 s.f.).

Q2. Round the given decimal numbers to the stated number of decimal places or significant figures: (i) 2.785 (2 d.p.), (ii) 88.9856 (4 s.f.), (iii) 15.3456 (2 d.p.), (iv) 10.4579 (3 s.f.).

(i) 2.785 to 2 d.p.: the third decimal digit is 5, so the second decimal digit 8 rounds up to 9, giving 2.79.

(ii) 88.9856 to 4 s.f.: the significant digits are 889856; keeping four gives 8899 (next digit 5 rounds the 8 up to 9), so 88.99.

(iii) 15.3456 to 2 d.p.: the third decimal digit is 5, so the second decimal digit 4 rounds up to 5, giving 15.35.

(iv) 10.4579 to 3 s.f.: the significant digits are 104579; keeping three gives 105 (next digit 5 rounds the 4 up to 5), so 10.5.

Q3. Round the integers away from zero (0) as given: (i) -23.456 (1 d.p.), (ii) -45325 (3 s.f.), (iii) -17.7898 (4 s.f.), (iv) 1172859 (5 s.f.).

Rule for negative numbers rounded away from zero: round the size (absolute value) up in the usual way, then put the minus sign back.

(i) |-23.456| = 23.456; to 1 d.p. the second decimal digit is 5, so 23.4 rounds up to 23.5, giving -23.5.

(ii) |-45325| = 45325; to 3 s.f. the first three digits are 453 and the next digit is 2, so it stays 453, giving -45300.

(iii) |-17.7898| = 17.7898; to 4 s.f. the digits are 1779 (next digit 9 rounds 8 up to 9), giving -17.79.

(iv) 1172859 to 5 s.f.: the first five digits are 11728 and the next digit is 5, so it rounds up to 11729, giving 1172900.

Q4. Round the rational numbers to the stated number of decimal places or significant figures: (i) 23/3 (3 d.p.), (ii) 55/3 (2 d.p.), (iii) 47/7 (2 d.p.), (iv) 22/6 (3 d.p.), (v) 25/29 (5 s.f.), (vi) 5/521 (4 s.f.).

Step 1: Change each fraction to a decimal first, then round it.

(i) 23/3 = 7.666...; to 3 d.p. the fourth decimal digit is 6, so 7.667.

(ii) 55/3 = 18.333...; to 2 d.p. the third decimal digit is 3, so 18.33.

(iii) 47/7 = 6.714285...; to 2 d.p. the third decimal digit is 4, so 6.71.

(iv) 22/6 = 3.666...; to 3 d.p. the fourth decimal digit is 6, so 3.667.

(v) 25/29 = 0.862068...; to 5 s.f. the digits are 86207 (next digit 6 rounds up), so 0.86207.

(vi) 5/521 = 0.0095969...; to 4 s.f. the significant digits are 9597 (next digit 6 rounds up), so 0.009597.

Q5. Write the number of significant figures in each of the following: (i) 0.0028, (ii) 25901, (iii) 0.1080, (iv) 0.000235.

Rule: Leading zeros are not significant; zeros between digits, and trailing zeros after a decimal point, are significant.

(i) 0.0028: the leading zeros are not counted, so only 2 and 8 count: 2 significant figures.

(ii) 25901: every digit counts, including the 0 between digits: 5 significant figures.

(iii) 0.1080: the leading zero is not counted, but 1, 0, 8 and the final 0 all count: 4 significant figures.

(iv) 0.000235: the leading zeros are not counted, so only 2, 3 and 5 count: 3 significant figures.

Q6. Compare your approximated value with the accurate value and also find an approximation error: (i) 3.78 / 1.98, (ii) 330 x 7.62 / 2.53, (iii) (28.35 + 30.15) / (4593 - 3690), (iv) (133 + 8.95) / 5.36.

Rule: Approximation error = |approximated value - accurate value|.

(i) Approximated value: 3.78 is about 4 and 1.98 is about 2, so 4 / 2 = 2.

Accurate value: 3.78 / 1.98 = 1.91 (approx.). Approximation error = |2 - 1.91| = 0.09.

(ii) Approximated value: 7.62 is about 8 and 2.53 is about 2.5, so 330 x 8 / 2.5 = 2640 / 2.5 = 1056.

Accurate value: 330 x 7.62 / 2.53 = 2514.6 / 2.53 = 993.91 (approx.).

Approximation error = |1056 - 993.91| = 62.09.

(iii) Approximated value: 28.35 + 30.15 is about 28 + 30 = 58, and 4593 - 3690 is about 4600 - 3700 = 900, so 58 / 900 = 0.064.

Accurate value: 58.50 / 903 = 0.065 (approx.). Approximation error = |0.064 - 0.065| = 0.001 (approx.).

(iv) Approximated value: 8.95 is about 9 and 5.36 is about 5, so (133 + 9) / 5 = 142 / 5 = 28.4.

Accurate value: 141.95 / 5.36 = 26.48 (approx.). Approximation error = |28.4 - 26.48| = 1.92.

Q7. The total length of the Great Wall of China is 21196.18 km. Round the number to 1 decimal place and 5 significant figures.

Step 1: To 1 decimal place, look at the second decimal digit, which is 8 (5 or more), so 21196.1 rounds up to 21196.2 km.

Step 2: To 5 significant figures, the digits are 21196 (the next digit is 1, so it stays unchanged), giving 21196 km.

Q8. The diameter of earth is 12742 km. Round the number to 3 and 4 significant figures.

Step 1: To 3 significant figures, the first three digits are 127 and the next digit is 4, so it stays 127, giving 12700 km.

Step 2: To 4 significant figures, the first four digits are 1274 and the next digit is 2, so it stays 1274, giving 12740 km.

Domain 1 — Numbers and Operations: Square Roots and Cube Roots

Exercise 1.4

Q1. Find the square of the following numbers: (i) 7, (ii) 11, (iii) 19, (iv) 25, (v) 37, (vi) 75.

Step 1: The square of a number is the number multiplied by itself.

(i) 7 x 7 = 49. (ii) 11 x 11 = 121. (iii) 19 x 19 = 361.

(iv) 25 x 25 = 625. (v) 37 x 37 = 1369. (vi) 75 x 75 = 5625.

Q2. Write the summation patterns for the following squares: (i) 6^2, (ii) 7^2, (iii) 4^2, (iv) 5^2, (v) 3^2, (vi) 8^2.

Rule: n^2 is the sum of the first n odd numbers, starting from 1.

(i) 6^2 = 1 + 3 + 5 + 7 + 9 + 11 = 36. (ii) 7^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 = 49.

(iii) 4^2 = 1 + 3 + 5 + 7 = 16. (iv) 5^2 = 1 + 3 + 5 + 7 + 9 = 25.

(v) 3^2 = 1 + 3 + 5 = 9. (vi) 8^2 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64.

Exercise 1.5

Q1. Find the square root of the following by prime factorization method: (i) 784, (ii) 1225, (iii) 2500, (iv) 4225, (v) 5184, (vi) 7744, (vii) 1296, (viii) 1764, (ix) 29241, (x) 51529, (xi) 418609, (xii) 240100.

Method: Factorise into primes, make pairs of equal primes, take one prime from each pair, and multiply them. Each answer below was checked by squaring it back.

(i) 784 = 2^4 x 7^2, so root = 2^2 x 7 = 28. (ii) 1225 = 5^2 x 7^2, so root = 5 x 7 = 35.

(iii) 2500 = 2^2 x 5^4, so root = 2 x 5^2 = 50. (iv) 4225 = 5^2 x 13^2, so root = 5 x 13 = 65.

(v) 5184 = 2^6 x 3^4, so root = 2^3 x 3^2 = 72. (vi) 7744 = 2^6 x 11^2, so root = 2^3 x 11 = 88.

(vii) 1296 = 2^4 x 3^4, so root = 2^2 x 3^2 = 36. (viii) 1764 = 2^2 x 3^2 x 7^2, so root = 2 x 3 x 7 = 42.

(ix) 29241 = 3^4 x 19^2, so root = 3^2 x 19 = 171. (x) 227^2 = 51529, so root = 227.

(xi) 647^2 = 418609, so root = 647. (xii) 490^2 = 240100, so root = 490.

Q2. Find the square root of the following by division method: (i) 1369, (ii) 13689, (iii) 50625, (iv) 103041.

Method: Pair the digits from the right, find the largest square at each step, double the quotient for the next divisor, and bring down the next pair.

(i) 1369: pairs 13, 69 give quotient 37, so root = 37 (37^2 = 1369).

(ii) 13689: pairs 1, 36, 89 give quotient 117, so root = 117 (117^2 = 13689).

(iii) 50625: pairs 5, 06, 25 give quotient 225, so root = 225 (225^2 = 50625).

(iv) 103041: pairs 10, 30, 41 give quotient 321, so root = 321 (321^2 = 103041).

Exercise 1.6

Q1. Find the square root of the following fractions by prime factorization: (i) 49/64, (ii) 121/625, (iii) 196/441, (iv) 1 13/36, (v) 676/729, (vi) 12 24/25.

Rule: The square root of a fraction is the square root of the numerator divided by the square root of the denominator.

(i) Root of 49/64 = 7/8. (ii) Root of 121/625 = 11/25. (iii) Root of 196/441 = 14/21 = 2/3.

(iv) 1 13/36 = 49/36, so its root = 7/6 = 1 1/6.

(v) Root of 676/729 = 26/27. (vi) 12 24/25 = 324/25, so its root = 18/5 = 3 3/5.

Q2. Find the square root of the following fractions by division method: (i) 144/225, (ii) 169/256, (iii) 5 41/64.

Step 1: Take the square root of the numerator and of the denominator separately by the division method.

(i) Root of 144/225 = 12/15 = 4/5. (ii) Root of 169/256 = 13/16.

(iii) 5 41/64 = 361/64, so its root = 19/8 = 2 3/8.

Exercise 1.7

Q1. Find the square root of the following decimals by prime factorization: (i) 1.21, (ii) 0.64, (iii) 7.29, (iv) 1.44, (v) 1.69, (vi) 12.25.

Method: Change the decimal to a fraction, take the square root of numerator and denominator, then change back to a decimal.

(i) 1.21 = 121/100, root = 11/10 = 1.1. (ii) 0.64 = 64/100, root = 8/10 = 0.8.

(iii) 7.29 = 729/100, root = 27/10 = 2.7. (iv) 1.44 = 144/100, root = 12/10 = 1.2.

(v) 1.69 = 169/100, root = 13/10 = 1.3. (vi) 12.25 = 1225/100, root = 35/10 = 3.5.

Q2. Find the square root of the following decimals by division method: (i) 0.3249, (ii) 0.5184, (iii) 10.24, (iv) 20.5209, (v) 3856.41, (vi) 1866.24, (vii) 0.680625, (viii) 0.797449, (ix) 1281.64.

Method: Pair the digits on both sides of the decimal point, then apply the division method as for whole numbers, placing the decimal point in the quotient at the same place.

(i) Root of 0.3249 = 0.57 (57^2 = 3249). (ii) Root of 0.5184 = 0.72 (72^2 = 5184).

(iii) Root of 10.24 = 3.2 (32^2 = 1024). (iv) Root of 20.5209 = 4.53 (453^2 = 205209).

(v) Root of 3856.41 = 62.1 (621^2 = 385641). (vi) Root of 1866.24 = 43.2 (432^2 = 186624).

(vii) Root of 0.680625 = 0.825 (825^2 = 680625). (viii) Root of 0.797449 = 0.893 (893^2 = 797449).

(ix) Root of 1281.64 = 35.8 (358^2 = 128164).

Exercise 1.8

Q1. Find the square root of the following up to 3 decimal places: (i) 2, (ii) 3, (iii) 5, (iv) 7, (v) 11, (vi) 21.

Method: Use the division method, continuing to three decimal places, then round the third decimal place.

(i) Root of 2 = 1.414 (approx.). (ii) Root of 3 = 1.732 (approx.). (iii) Root of 5 = 2.236 (approx.).

(iv) Root of 7 = 2.646 (approx.). (v) Root of 11 = 3.317 (approx.). (vi) Root of 21 = 4.583 (approx.).

Q2. Find the square root of the following up to 2 decimal places: (i) 3.6, (ii) 6.4, (iii) 28.9, (iv) 63.34, (v) 921.25, (vi) 5921.25.

Method: Use the division method to three decimal places first, then round to two decimal places.

(i) Root of 3.6 = 1.897..., so 1.90 (2 d.p.). (ii) Root of 6.4 = 2.529..., so 2.53 (2 d.p.).

(iii) Root of 28.9 = 5.376..., so 5.38 (2 d.p.). (iv) Root of 63.34 = 7.958..., so 7.96 (2 d.p.).

(v) Root of 921.25 = 30.354..., so 30.35 (2 d.p.). (vi) Root of 5921.25 = 76.952..., so 76.95 (2 d.p.).

Exercise 1.9

Q1. Find the number of digits in the square root of the following perfect square and verify it by finding square root: (i) 63504, (ii) 66564, (iii) 50625, (iv) 837225, (v) 839056, (vi) 1054729, (vii) 1577536, (viii) 2119936, (ix) 3283344, (x) 614656, (xi) 7778521, (xii) 12880921.

Rule: If n is the number of digits in the perfect square, its root has n/2 digits if n is even, and (n+1)/2 digits if n is odd.

(i) 63504 has 5 digits (odd), so its root has (5+1)/2 = 3 digits; root = 252, which indeed has 3 digits.

(ii) 66564 has 5 digits, so its root has 3 digits; root = 258. (iii) 50625 has 5 digits, so its root has 3 digits; root = 225.

(iv) 837225 has 6 digits (even), so its root has 6/2 = 3 digits; root = 915.

(v) 839056 has 6 digits, so its root has 3 digits; root = 916. (vi) 1054729 has 7 digits, so its root has 4 digits; root = 1027.

(vii) 1577536 has 7 digits, so its root has 4 digits; root = 1256. (viii) 2119936 has 7 digits, so its root has 4 digits; root = 1456.

(ix) 3283344 has 7 digits, so its root has 4 digits; root = 1812. (x) 614656 has 6 digits, so its root has 3 digits; root = 784.

(xi) 7778521 has 7 digits, so its root has 4 digits; root = 2789. (xii) 12880921 has 8 digits, so its root has 4 digits; root = 3589.

Exercise 1.10

Q1. The length of a squared shaped box is 324 cm. Find the area of the box.

Step 1: The box is square, so its area = side x side.

Step 2: Area = 324 x 324 = 104976 square cm.

Q2. The length of a square shaped field is 36 m. Find the area of the square field.

Step 1: Area of a square = side x side.

Step 2: Area = 36 x 36 = 1296 square metres.

Q3. The length of a square shaped lawn is 250 metres. What will be the area of the lawn and also tell what will be its cost of growing grass at the rate of Rs. 75 per sq. metres?

Step 1: Area = 250 x 250 = 62500 square metres.

Step 2: Cost of grass = area x rate = 62500 x 75 = Rs. 4687500.

Q4. The area of a square shaped field is 14400 sq. metres. Find the length of the side of the square.

Step 1: Side = square root of area.

Step 2: Side = root of 14400 = 120 metres, because 120 x 120 = 14400.

Q5. The area of a square shaped field is 422500 sq. metres. How much string is required for fixing along the sides as a fence?

Step 1: Side = root of 422500 = 650 metres, because 650 x 650 = 422500.

Step 2: Fence goes all round, so string needed = perimeter = 4 x 650 = 2600 metres.

Q6. A gardener wants to plant 122500 trees in his field in such a way that the number of trees in a row is equal to the number of rows. How many trees will he plant in each row?

Step 1: Trees in each row = square root of the total number of trees.

Step 2: Root of 122500 = 350, because 350 x 350 = 122500, so 350 trees in each row.

Q7. The area of a rectangular field is 10092 sq. metres. Its length is three times as long as its width. Find its perimeter.

Step 1: Let the width be w metres, so the length is 3w metres.

Step 2: Area = length x width = 3w x w = 3w^2 = 10092, so w^2 = 10092 / 3 = 3364.

Step 3: Width w = root of 3364 = 58 metres, so length = 3 x 58 = 174 metres.

Step 4: Perimeter = 2 x (length + width) = 2 x (174 + 58) = 2 x 232 = 464 metres.

Q8. A rectangular field has an area 28800 sq. metres. Its length is twice as long as its width. What is the length of its sides?

Step 1: Let the width be w metres, so the length is 2w metres.

Step 2: Area = 2w x w = 2w^2 = 28800, so w^2 = 28800 / 2 = 14400.

Step 3: Width w = root of 14400 = 120 metres, so length = 2 x 120 = 240 metres.

Q9. Find that least number which, when subtracted from 109087, the answer is a complete square.

Step 1: Find the square root of 109087 by division: the largest whole root is 330, because 330^2 = 108900.

Step 2: The remainder is 109087 - 108900 = 187.

Step 3: So 187 is the least number to subtract; 109087 - 187 = 108900 = 330^2, a complete square.

Q10. The cost of levelling the ground of a circular region at a rate of Rs. 2 per square metre is Rs. 4928. Find the radius of the ground.

Step 1: Area of the ground = total cost / rate = 4928 / 2 = 2464 square metres.

Step 2: Area of a circle = pi x r^2, using pi = 22/7, so r^2 = 2464 x 7 / 22 = 784.

Step 3: Radius r = root of 784 = 28 metres.

Q11. The cost of ploughing in a square field is Rs. 2450 at the rate of Rs. 2 per 100 sq. metres. Find the length of the side of the square.

Step 1: Area of the field = 2450 / 2 x 100 = 122500 square metres.

Step 2: Side = root of 122500 = 350 metres, because 350 x 350 = 122500.

Q12. The area of a square shaped lawn is 62500 sq. metres. A wooden fence is to be laid around the lawn. What is the length of a wooden fence required? What will be its cost at the rate of Rs. 50 per metre?

Step 1: Side = root of 62500 = 250 metres, because 250 x 250 = 62500.

Step 2: Fence length = perimeter = 4 x 250 = 1000 metres.

Step 3: Cost = 1000 x 50 = Rs. 50000.

Exercise 1.11

Q1. Which numbers are perfect cubes? (i) 512, (ii) 1100, (iii) 6859, (iv) 729, (v) 1000/125.

Step 1: A perfect cube is the cube of a whole number.

(i) 512 = 8 x 8 x 8 = 8^3, so it is a perfect cube. (ii) 1100 is not the cube of any whole number (10^3 = 1000, 11^3 = 1331), so it is not a perfect cube.

(iii) 6859 = 19 x 19 x 19 = 19^3, so it is a perfect cube. (iv) 729 = 9 x 9 x 9 = 9^3, so it is a perfect cube.

(v) 1000/125 = 8 = 2 x 2 x 2 = 2^3, so it is a perfect cube.

Q2. Find the cube roots of the following: (i) 729, (ii) 15625, (iii) 13824.

Step 1: Factorise each number into three equal groups of prime factors, or check by cubing.

(i) Cube root of 729 = 9, because 9^3 = 729. (ii) Cube root of 15625 = 25, because 25^3 = 15625.

(iii) Cube root of 13824 = 24, because 24^3 = 13824.

Q3. Find the cubes of the following: (i) 1.4, (ii) 0.4, (iii) 0.8.

Step 1: Cube means multiply the number by itself three times.

(i) 1.4^3 = 1.4 x 1.4 x 1.4 = 2.744. (ii) 0.4^3 = 0.4 x 0.4 x 0.4 = 0.064.

(iii) 0.8^3 = 0.8 x 0.8 x 0.8 = 0.512.

Q4. Find the cube roots of the following: (i) 27/216, (ii) 35937, (iii) 3375.

Step 1: For a fraction, take the cube root of the numerator and of the denominator separately.

(i) Cube root of 27/216 = 3/6 = 1/2. (ii) Cube root of 35937 = 33, because 33^3 = 35937.

(iii) Cube root of 3375 = 15, because 15^3 = 3375.

Q5. If a cube has length of 6 cm, then what will be the volume of the cube?

Step 1: Volume of a cube = side x side x side.

Step 2: Volume = 6 x 6 x 6 = 216 cubic cm.

Q6. Find the volume of a cube if the length of one side is 8 cm.

Step 1: Volume = side^3.

Step 2: Volume = 8 x 8 x 8 = 512 cubic cm.

Q7. If the volume of a cube shape is 3375 cm^3, then what is the measure of the edge of the cube?

Step 1: Edge = cube root of volume.

Step 2: Edge = cube root of 3375 = 15 cm, because 15^3 = 3375.

Q8. The volume of a cube is 729 m^3. What will be the length of a side of the cube?

Step 1: Side = cube root of volume.

Step 2: Side = cube root of 729 = 9 m, because 9^3 = 729.

Q9. How many cubical boxes of dimensions 5 cm x 5 cm x 5 cm can be packed in a large cubical case having volume 875 cm^3?

Step 1: Volume of one small box = 5 x 5 x 5 = 125 cubic cm.

Step 2: Number of boxes = total volume / volume of one box = 875 / 125 = 7 boxes.

Review Exercise 1 (a)

Q1. Four options are given against each statement. Encircle the correct one. (i) Real number is: (a) Difference of rational and irrational numbers, (b) Intersection of rational and irrational numbers, (c) Union of rational and irrational numbers, (d) Complete set of natural numbers. (ii) Which of the following is true about root 8? (a) Natural, (b) Whole, (c) Rational, (d) Irrational. (iii) Irrational numbers are non terminating and ____ numbers: (a) rational, (b) non-recurring, (c) recurring, (d) decimal. (iv) Round 0.0234589

to 4 significant figures: (a) 0.02346, (b) 0.02345, (c) 2346, (d) 0.0235. (v) The difference between approximated value and actual accurate value is called: (a) round, (b) significant figure, (c) approximation error, (d) estimated value. (vi) Which one of the following is a perfect square? (a) 25.6, (b) 0.256, (c) 2.56, (d) 2560. (vii) Square of 0.9 is: (a) 0.081, (b) 8.10, (c) 0.81, (d) 81.027. (viii) (7/9)^2 = ? (a) 49/6, (b) 7/81, (c) 49/81, (d) 7/3. (ix) 1+2+3+4+5+6+7+8+7+6+5+4+3+2+1 = ? (a) 8^2, (b) 9^2, (c) 65, (d) 81. (x) If the side length of a square is 0.5 m, then its area is: (a) 0.50 m^2, (b) 2.5 m^2, (c) 0.25 m^2, (d) 25 m^2. (xi) Root of 0.04 = ? (a) 0.02, (b) 2.0, (c) 0.2, (d) 20. (xii) Root of (1^2 x 4^2) = ? (a) 4, (b) 14, (c) 41, (d) 2. (xiii) Root of (4/9) = ? (a) 2/3, (b) 2/9, (c) 4/3, (d) 3/2. (xiv) Root of (a/b) = ? (a) a/b, (b) ab, (c) root(b/a), (d) root(a)/root(b).

(i) Correct option: (c) Union of rational and irrational numbers, because every real number is either rational or irrational, and together they make up all real numbers. **Correct: (c)**

(ii) Correct option: (d) Irrational number, because 8 is not a perfect square, so its square root cannot be written as a fraction. **Correct: (d)**

(iii) Correct option: (b) non-recurring, because irrational decimals never terminate and never repeat a block of digits. **Correct: (b)**

(iv) Correct option: (a) 0.02346, because the significant digits are 234589 and keeping four gives 2346 (the next digit 8 rounds the 5 up to 6). **Correct: (a)**

(v) Correct option: (c) approximation error, which is the difference between the approximated value and the accurate value. **Correct: (c)**

(vi) Correct option: (c) 2.56, because 1.6 x 1.6 = 2.56, so 2.56 is the square of 1.6. **Correct: (c)**

(vii) Correct option: (c) 0.81, because 0.9 x 0.9 = 0.81. **Correct: (c)**

(viii) Correct option: (c) 49/81, because (7/9)^2 = 7^2 / 9^2 = 49/81.

Correct: (c)

(ix) Correct option: (a) 8^2, because the sum rises 1 to 8 and falls back to 1, which is the summation pattern for 8^2 = 64. **Correct: (a)**

(x) Correct option: (c) 0.25 m^2, because area = 0.5 x 0.5 = 0.25 square metres. **Correct: (c)**

(xi) Correct option: (c) 0.2, because 0.2 x 0.2 = 0.04. **Correct: (c)**

(xii) Correct option: (a) 4, because root of (1^2 x 4^2) = 1 x 4 = 4. **Correct: (a)**

(xiii) Correct option: (a) 2/3, because root of (4/9) = 2/3. **Correct: (a)**

(xiv) Correct option: (d) root(a)/root(b), because the square root of a fraction is the square root of the numerator divided by the square root of the denominator. **Correct: (d)**

Q2. Identify terminating, non-terminating and recurring decimal fractions from the following: (i) 17/5, (ii) 23/3, (iii) 49/8, (iv) 8/15.

Step 1: Change each fraction to a decimal and look at whether it ends or repeats.

(i) 17/5 = 3.4, which ends, so it is a terminating decimal fraction.

(ii) 23/3 = 7.666..., which never ends and repeats the digit 6, so it is a non-terminating recurring decimal fraction.

(iii) 49/8 = 6.125, which ends, so it is a terminating decimal fraction.

(iv) 8/15 = 0.5333..., which never ends and repeats the digit 3, so it is a non-terminating recurring decimal fraction.

Q3. Separate rational and irrational numbers: (i) 25/6, (ii) 55/7, (iii) 8.111111, (iv) root 21.

Step 1: A number is rational if it can be written as a fraction, or as a terminating or recurring decimal.

(i) 25/6 is a fraction, so it is rational (its decimal 4.1666... recurs).

(ii) 55/7 is a fraction, so it is rational (its decimal 7.857142... recurs).

(iii) 8.111111 has a fixed number of digits, so it is a terminating decimal and therefore rational.

(iv) Root 21 is irrational, because 21 is not a perfect square, so its root cannot be written as a fraction.

Q4. Identify the property of real numbers used in the following: (i) 25 + (7 + 5) = (25 + 7) + 5, (ii) 17 + (-17) = 0, (iii) root z + 0 = root z, (iv) 25 x 5 = 5 x 25.

(i) Associative property of addition, because only the grouping changes and the sum stays the same.

(ii) Additive inverse, because a number plus its negative gives 0.

(iii) Additive identity, because adding 0 leaves a number unchanged.

(iv) Commutative property of multiplication, because only the order changes and the product stays the same.

Q5. Round the following to the given degree of accuracy: (i) 598235 (4 s.f.), (ii) 0.002859 (3 s.f.), (iii) 45.35946 (2 d.p.), (iv) 25/8 (2 d.p.), (v) 15/7 (2 d.p.), (vi) 0.078952 (4 s.f.).

(i) 598235 to 4 s.f.: the first four digits are 5982 and the next digit is 3, so it stays 5982, giving 598200.

(ii) 0.002859 to 3 s.f.: the significant digits are 2859; keeping three gives 286 (next digit 9 rounds up), so 0.00286.

(iii) 45.35946 to 2 d.p.: the third decimal digit is 9, so the second decimal digit 5 rounds up to 6, giving 45.36.

(iv) 25/8 = 3.125; to 2 d.p. the third decimal digit is 5, so the second decimal digit 2 rounds up to 3, giving 3.13.

(v) 15/7 = 2.142857...; to 2 d.p. the third decimal digit is 2, so it stays 2.14.

(vi) 0.078952 to 4 s.f.: the significant digits are 78952; keeping four gives 7895 (next digit 2 leaves it unchanged), so 0.07895.

Q6. Javeria bought 25.75 metres of cloth and Salma bought 78.28 metres of cloth. Find the total length of cloth. Also round the answer too: (i) 3 s.f., (ii) 1 d.p.

Step 1: Total length = 25.75 + 78.28 = 104.03 metres.

Step 2: (i) To 3 significant figures, the digits are 104 (the next digit is 0), giving 104 metres.

Step 3: (ii) To 1 decimal place, the second decimal digit is 3, so it stays 104.0 metres.

Q7. The cost of 10 kg rice is Rs. 1557 and the cost of 15 kg sugar is Rs. 1278. What will be cost of both the items. Also round the answer up to 2 significant figures.

Step 1: Total cost = 1557 + 1278 = Rs. 2835.

Step 2: To 2 significant figures, the first two digits are 28 and the next digit is 3, so it stays 28, giving Rs. 2800.

Q8. Find the number of digits in the square root of the following numbers. Also find the square root: (i) 9604, (ii) 30625, (iii) 12544, (iv) 422500.

Rule: If n is the number of digits in the perfect square, its root has n/2 digits if n is even, and (n+1)/2 digits if n is odd.

(i) 9604 has 4 digits (even), so its root has 2 digits; root = 98, because 98^2 = 9604.

(ii) 30625 has 5 digits (odd), so its root has 3 digits; root = 175, because 175^2 = 30625.

(iii) 12544 has 5 digits, so its root has 3 digits; root = 112, because 112^2 = 12544.

(iv) 422500 has 6 digits (even), so its root has 3 digits; root = 650, because 650^2 = 422500.

Q9. Find the square root of the following: (i) 28 4/9, (ii) 17 128/289, (iii) 101 92/169, (iv) 0.053361, (v) 0.204304, (vi) 152.7696, (vii) 0.316406, (viii) 38.01 (2 d.p.), (ix) 6422.53 (2 d.p.).

(i) 28 4/9 = 256/9, so its root = 16/3 = 5 1/3.

(ii) 17 128/289 = 5041/289, so its root = 71/17 = 4 3/17.

(iii) 101 92/169 = 17161/169, so its root = 131/13 = 10 1/13.

(iv) Root of 0.053361 = 0.231, because 231^2 = 53361.

(v) Root of 0.204304 = 0.452, because 452^2 = 204304.

(vi) Root of 152.7696 = 12.36, because 1236^2 = 1527696.

(vii) Root of 0.316406 = 0.5625 (approx., because 0.5625^2 = 0.31640625).

(viii) Root of 38.01 = 6.165..., so 6.17 (2 d.p.).

(ix) Root of 6422.53 = 80.140..., so 80.14 (2 d.p.).

Q10. If the area of a square field is 161604 m^2. Find the length of its one side.

Step 1: Side = square root of area.

Step 2: Side = root of 161604 = 402 metres, because 402 x 402 = 161604.

Q11. Saeeda has 196 marbles that she is using to make a square formation. How many marbles should be in each row?

Step 1: Marbles in each row = square root of the total number of marbles.

Step 2: Root of 196 = 14, because 14 x 14 = 196, so 14 marbles in each row.

Q12. Find the cube root of the following numbers: (i) 1728, (ii) 3375, (iii) 216/125.

(i) Cube root of 1728 = 12, because 12^3 = 1728.

(ii) Cube root of 3375 = 15, because 15^3 = 3375.

(iii) Cube root of 216/125 = 6/5, because 6^3 = 216 and 5^3 = 125.

Q13. The volume of a cube is 512 mm^3. What will be the length of a side of the cube?

Step 1: Side = cube root of volume.

Step 2: Side = cube root of 512 = 8 mm, because 8^3 = 512.

Q14. What will be the volume of cube, whose side is 10 cm long?

Step 1: Volume = side x side x side.

Step 2: Volume = 10 x 10 x 10 = 1000 cubic cm.